

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: 25UD1293PE504C - SMART GRID TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which organization promotes Smart Grid activities in India?
a) RBI b) ISRO c) Ministry of Power d) DRDO
- 2 Which Indian initiative focuses on Smart Grid pilot projects?
a) NSGM b) UDAY c) SAUBHAGYA d) DDUGJY
- 3 Smart Grid improves power quality by:
a) Increasing faults b) Faster fault detection
c) Increasing outages d) Reducing automation
- 4 Smart Grid architecture mainly consists of:
a) Single layer b) Control, communication, power layers
c) Mechanical layer only d) Market layer only
- 5 IP-based systems in Smart Grid allow:
a) Isolated operation b) Internet-based communication
c) Manual control d) Fixed topology
- 6 Which standard is widely used for Smart Grid interoperability?
a) IEEE b) BIS 10500 c) ISO 9001 d) IEC 61850
- 7 Dynamic optimization deals with:
a) Steady state only b) Manual scheduling
c) Time-dependent behavior d) Fixed load
- 8 Artificial Neural Networks are inspired by:
a) Human brain b) Electrical circuits c) Transformers d) Motors
- 9 A microgrid can operate in:
a) Grid-connected mode only b) Islanded mode only
c) Both modes d) Manual mode
- 10 Economic issues of DG include:
a) Capital cost b) Maintenance cost
c) Tariff structure d) All of the above

- 11 PMUs are used for:
 a) Billing b) Synchrophasor measurement
 c) Load shedding d) Meter testing
- 12 Smart cities focus on:
 a) Energy, transport, water b) Only buildings
 c) Only roads d) Only power plants
- Q2. Solve the following.
- A) Explain the National Smart Grid Mission (NSGM) launched by the Ministry of Power, Government of India. Discuss its objectives and key milestones with a neat diagram. 6
- B) Define Smart Grid. Explain how a Smart Grid differs from a conventional power grid. 6
- Q3. Solve the following.
- A) Discuss the major domains of a Smart Grid with a neat block diagram. 6
- B) Explain the importance of static and dynamic optimization techniques in Smart Grids. 6
- Q4. Solve Any Two of the following.
- A) Write short notes on: (a) SCADA (b) Advanced Metering Infrastructure (AMI). 6
- B) Discuss the application of Economic Load Dispatch (ELD) in Smart Grids. 6
- C) Explain various renewable energy technologies used in Smart Grids. 6
- Q5. Solve Any Two of the following.
- A) Explain the concept of Distributed Generation (DG) and discuss its necessity in modernizing the conventional power grid. Elaborate on how DG supports bidirectional power flow in Smart Grids and reduces transmission losses. 6
- B) Explain the crucial role of energy storage technologies in the context of Distributed Generation and Smart Grids. Illustrate how storage systems are used for functions like smoothing renewable variability and peak shaving to balance supply and demand. 6
- C) How do Electric Vehicles (EVs) and Plug-in Hybrid Electric Vehicles (PHEVs) function as both mobile loads and potential distributed storage? Explain the Vehicle-to-Grid (V2G) concept and its potential benefits for grid flexibility. 6
- Q6. Solve Any Two of the following.
- A) Explain how intelligent techniques can be applied for power system operation. 6
- B) Explain the role of the Internet of Things (IoT) in the Smart Grid ecosystem. Give three key applications of IoT, such as smart meters, load forecasting, or grid automation, that facilitate data-driven decision-making and remote control. 6
- C) Discuss the relationship between Smart Grids and Smart Cities. Explain the concept of Active Distribution Networks (ADN) and how they utilize real-time control of distributed energy sources to improve voltage regulation and fault restoration. 6

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